Prashast Awasthi

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Portfolio: awasthiprashast.github.io

Education

Vellore Institute of Technology Vellore, India
B.Tech in Computer Science and Engineering — CGPA: 8.09 / 10.0 *Graduating: 2027*

Technical Skills

Languages: Python, SQL, C++
Machine Learning: scikit-learn, Regression, Classification, Random Forest, XGBoost, LightGBM, Time-Series Forecasting (SARIMA, Prophet), Model Evaluation, SHAP
Data Science & Analytics: Pandas, NumPy, EDA, Statistical Analysis, Data Cleaning, Data Visualization (Plotly, Matplotlib), Power BI
Deep Learning, NLP & CV: Hugging Face Transformers, FinBERT, Embeddings, OpenCV, MediaPipe, InsightFace
Generative AI: LLMs, RAG, AI Agents, Multi-Agent Systems, LangGraph, Prompt Engineering, ChromaDB, FAISS
Tools & Deployment: PostgreSQL, MongoDB, FastAPI, Streamlit, Docker, Git/GitHub, pytest

Experience

Neve Jewels Mumbai, India
Data Science Intern *May 2026 – July 2026*

- Built next-day demand forecasting models for 3 jewellery brands on 2.5 years of daily order data; XGBoost achieved **18.7% MAPE** under walk-forward cross-validation, outperforming 5 other models incl. LightGBM, SARIMA and Prophet.
- Diagnosed a silent data-quality bug: a left join with campaign data followed by dropna deleted every non-promotional day, so models trained only on promotion periods; fixing it restored the full history and removed the bias.
- Engineered **25+ time-series features** (lags, rolling statistics, calendar, discount, marketing); ran ablation studies and kept a feature group only if it reduced validation error.
- Applied SHAP for model explainability, identifying lagged orders and discount depth as the top demand drivers, and presented insights to business stakeholders in an interactive Plotly report.

AceQuant Research - Remote Mumbai, India
Quantitative Development Intern *May 2025 – July 2025*

- Implemented 4 trading strategies (SMA/EMA crossovers, Bollinger Bands, MACD) in Python, computing technical indicators and signals with NumPy on historical and live market data.
- Integrated Angel One and Zerodha API data into a backtesting workflow, evaluating parameter-tuned strategy variants on **6+ performance metrics**, incl. profitability and drawdown.

Projects

Multi-Agent Financial Research Assistant | *Python, LLMs, RAG, FinBERT, ChromaDB, FastAPI*

- Built an 8-agent LLM equity research system: an orchestrator routes queries to technical, fundamental, risk and sentiment agents, retrieves context via RAG over ChromaDB and merges outputs into one report.
- Applied FinBERT sentiment analysis to **5,000+ news articles** and measured its correlation with price moves across 10+ equities.
- Designed a verification agent to guard against LLM hallucinations: it recomputes every numeric claim from source data and rejects results beyond 0.5% tolerance via a bounded retry loop; 22 modules, pytest-tested.

Face Recognition Photo Retrieval System | *Python, InsightFace, FAISS, FastAPI, React*

- Developed a face recognition and retrieval system that indexes InsightFace embeddings in FAISS for vector similarity search, returning each guest's photos from a multi-thousand-image event archive with **300+ guests**.
- Tuned the cosine-distance threshold to balance precision and recall; served results through a FastAPI backend, grouped by sub-event.

Computer Vision-Based Attention Monitoring System | *Python, OpenCV, MediaPipe, Streamlit*

- Engineered Eye Aspect Ratio (EAR) and head-yaw (PnP pose estimation) features from MediaPipe FaceMesh landmarks to detect eye closure and looking away.
- Aggregated frame-level signals into a time-series attention score, visualized on a real-time Streamlit dashboard.

Leadership & Activities

IEEE Professional Communication Society (PCS) – Vice Chairperson VIT University
March 2025 – March 2026

- Directed chapter operations and recruitment workflows, sustaining a 40% increase in active student engagement.
- Coordinated 3 cross-functional teams spanning Design, Technical, and Management to organize and deliver technical and cultural events; mentored 30+ junior members in technical and organizational activities.

Certifications

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional (Oracle University) *Aug 2026*